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Hydraulic conductivity reduction can be decoupled from plant available water and can influence nitrogen leaching in amended sandy soils

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Improvements in sandy-soil hydraulic behavior are often inferred from reduced permeability rather than from direct gains in plant-available water (AW). This study quantified the extent to which amendment-induced reductions in saturated hydraulic conductivity (K_s) correspond to changes in AW and how these responses influence nitrogen (N) leaching. Controlled soil-column experiments evaluated the effects of clay powder, compost, and a biochar-based biofertilizer (biofertilizer-char) on soil water retention characteristics, K_s , water discharge, and N-losses. Clay powder significantly reduced K_s but did not significantly increase field capacity, permanent wilting point, or AW, demonstrating that restricted drainage does not necessarily improve water availability. Compost increased saturation percentage (18.6–20.8%), but higher application rates reduced AW (10.4–7.6%) and increased N-leaching. Combined compost and biofertilizer-char treatments maintained AW (~10.3%) while reducing water discharge and moderating N-losses. Nitrogen leaching varied among amendment treatments. High compost rates produced the greatest losses (0.63 mg column⁻¹, 15.75%), while moderate compost and compost + biofertilizer-char treatments reduced losses to 0.09–0.20 mg column⁻¹ (3.25–8.67%). Leached-N increased with initial total N-content ($R^2 = 0.378$) and decreased with leachate volume ($R^2 = 0.253$). These findings demonstrate that reduction in K_s can be decoupled from gains in AW and that N-leaching is jointly regulated by nitrogen availability and amendment-induced modifications of soil water transport.

Keywords Soil physical modification, Organic amendments, Biochar-compost interaction, Clay powder

Nitrogen loss through leaching represents a major pathway of nutrient inefficiency and environmental contamination in coarse-textured soils, particularly in arid and semi-arid regions where water management is tightly coupled to soil physical properties^{1–3}. Sandy soils, which dominate large areas of these regions, are characterized by a prevalence of macropores, low colloid content, and weak structural development, resulting in rapid water transmission, limited water retention, and enhance solute transport^{4–8}. These physical constraints reduce water-use efficiency and promote nitrogen losses, often requiring more intensive irrigation and fertilizer management to sustain crop productivity. In sandy soils, high-frequency irrigation combined with fertigation can partially alleviate these limitations by improving water and nutrient use efficiency^{9–11}. However, these management practices do not directly improve the soil's inherently low water- and nutrient-retention capacity. Therefore, mineral and organic amendments are widely applied to improve the physical functioning of sandy soils and complement efficient irrigation management. Clay minerals are commonly incorporated to increase fine particle content and restrict rapid drainage^{12,13}, while organic amendments such as compost and biochar are used to enhance aggregation, alter pore-size distribution, and increase soil water retention^{14–17}. These

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approaches are generally based on the assumption that reducing permeability and increasing water retention will improve plant-available water and limit nutrient losses. Unlike mineral amendments that primarily restrict water movement through pore-size reduction, biochar-based amendments may alter soil hydraulic behavior through multiple pathways, including changes in pore architecture, aggregation, and microbial activity¹⁸. Consequently, their net effect on hydraulic conductivity and water retention is context dependent and requires experimental evaluation.

However, this assumption overlooks a critical distinction between hydraulic restriction and effective water availability. Water retained at high matric suction or confined within fine pores may be inaccessible to plants, meaning that reduction in permeability or increases in total water content do not necessarily translate into gains in plant-available water^{19–21}. In sandy soils, where slight changes in pore-size distribution can substantially alter water retention characteristics, this distinction becomes particularly important. A reduction in drainage may coincide with an increase in water held beyond the range accessible for plant uptake, effectively decoupling hydraulic conductivity from functional water availability. Despite being well recognized conceptually, the extent to which amendment-induced reductions in saturated hydraulic conductivity are accompanied by proportional changes in plant-available water has rarely been quantified explicitly within integrated experimental systems that simultaneously assess hydraulic behavior and solute transport.

Nitrogen dynamics in sandy soils further depend on this combination between water flow and soil physical structure. Due to low sorption capacity and rapid water movement, urea-derived nitrogen is highly susceptible to leaching losses^{22,23}. Organic amendments are often promoted as strategies to reduce nitrogen losses through enhanced retention, microbial immobilization, or adsorption processes^{24–27}. However, increased nitrogen availability or amendment-induced changes in pore architecture may also enhance leaching under certain conditions^{28,29}. Consequently, nitrogen losses cannot be explained solely by fertilizer input but must be interpreted as an emergent outcome of interactions between soil physical properties, water transport, and nitrogen availability.

Despite extensive research on individual amendments, integrated experimental assessments that simultaneously quantify hydraulic conductivity, soil water retention, water discharge, and nitrogen leaching within the same experimental system remain limited. Moreover, few studies explicitly distinguish between total water retention and plant-available water while linking these properties to solute transport. This gap constrains the development of amendment strategies that effectively balance water retention with controlled drainage and nutrient conservation.

Although the effect of soil amendment on crop performance under water limited field conditions have been previously investigated³⁰, the underlying physical mechanisms through which these amendments alter water flow and nitrogen transport remain inadequately resolved. To address this knowledge gap, the present study evaluated the following null hypotheses under controlled soil column conditions: (H0₁) amendment-induced reductions in saturated hydraulic conductivity by soil amendments are accompanied by proportional increases in plant available water, (H0₂) organic and inorganic amendments applied at equivalent rates exert comparable effects on water discharge and nitrogen leaching, and (H0₃) nitrogen leaching is determined primarily by fertilizer application rate rather than by amendment-induced modifications of soil physical properties and flow dynamics. Rejection of these hypotheses would demonstrate that amendment effects are mechanistically distinct and that nitrogen losses are governed not only by fertilizer inputs but also by changes in soil hydraulic behavior and associated hydrological transport processes.

To address these hypotheses, a controlled soil column experiment was conducted to evaluate the effects of clay powder, compost, a biochar-based biofertilizer, and their combinations on the physical properties of sandy soil and the resulting water and nitrogen transport dynamics. The study was conducted in two sequential experiments. Experiment I compared mineral (clay powder) and organic amendments for their effects on soil physical properties. Experiment II focused on nitrogen leaching and included only those amendments selected based on their performance in Experiment I.

By quantitatively evaluating the relationships among amendment-induced changes in saturated hydraulic conductivity, plant available water, water discharge, and nitrogen movement, this study provides a mechanistic understanding of water-solute interactions in amended sandy soils. Rather than treating reduced permeability and improved water availability as equivalent outcomes, the study explicitly examines the extent to which these responses diverge across mineral and organic amendment systems and how this divergence influences nitrogen transport.

Material and methods

Soil sampling and preparation

Sandy soil samples were collected from a uniform sandy landscape area in non-protected land within the vicinity of the Faculty of Agriculture and Natural Resources, Aswan University, Aswan Governorate, Egypt (24° 05' 20.18" N, 32° 53' 59.39" E). No permits were required for sample collection, and the study did not involve protected areas, endangered species, or regulated sites. The average monthly air temperature and total reference evapotranspiration of the location is provided in the Supplementary Material (Fig. S1). All experimental work was conducted under controlled laboratory conditions. To account for small-scale spatial variability within the study area, soil was collected from three locations separated by approximately 5–6 m. Samples were collected from 0 to 30 cm surface layer where the majority of root biomass is concentrated^{31,32}. The collected soil was composited and homogenized prior to laboratory analysis and experimental use. Soil aggregates were air-dried, gently crushed, and passed through a 2-mm sieve prior to analysis and experimentation to ensure homogeneous packing among experimental units. The study site represents a coarse-textured sandy soil typical of many newly reclaimed agricultural areas in arid regions, where low water-holding capacity and high susceptibility to nutrient leaching are common constraints. Particle-size distribution was determined using the pipette method³³.

Soil amendments

Three soil amendments were evaluated: clay powder (CP), compost (C), and biochar-based biofertilizer (hereafter referred to as biofertilizer-char or B). Clay powder and compost were commercially available in Egypt, while biofertilizer char was prepared under laboratory conditions.

Clay powder mineralogy was determined by X-ray diffraction (XRD) analysis. The compost was obtained from Nile Commercial Company (Egypt). The biofertilizer-char was supplied by the Genetics Department, Faculty of Agriculture, Beni-Suef University. The biofertilizer-char consisted of biochar used as a carrier for two plant growth-promoting rhizobacteria, *Azotobacter sp.* and *Asospirillum sp.*, which are known for atmospheric nitrogen fixation, phosphate solubilization, and indole-3-acetic acid production.

Chemical characterization of the soil amendments included determination of pH, electrical conductivity (EC), calcium carbonate content, soluble cations and anions, and nutrients concentrations, following the standard procedures described by³⁴. Loss-on-ignition (LOI) method was used to estimate organic matter content. Samples were oven-dried at 105 °C for 24 h prior ignition at 360 °C for 3 h following the procedure described by³⁵. Total nitrogen content of the soil, compost, and biofertilizer-char was determined prior to column preparation using the Kjeldahl digestion method³⁶. Total nitrogen content per column (mg N Kg⁻¹) was calculated from the measured nitrogen concentrations of the soil and amendments together with the amount of urea-N applied to each treatment. These values represent the initial nitrogen concentration available within each column before leaching experiment. The detailed chemical properties of sand samples are provided in Tables S1 of the Supplementary Materials. The chemical characteristics for the used natural-based soil amendments (compost and biofertilizer-char) are presented in Table 1.

Experiment I: effect of soil amendments on soil physical properties

Experimental design

A laboratory soil column experiment was conducted to evaluate the effects of soil amendments on key physical properties of sandy soil. Transparent acrylic columns (4 cm internal diameter, 15 cm height) were used. One end of each column was sealed with medium speed filter paper (No. 102) supported by a layer of medical cotton gauze to prevent soil loss while allowing free drainage. A simplified diagram of the experiment is illustrated in Supplementary Materials (Fig. S2.)

Amendments were air-dried and thoroughly mixed with a fixed mass (150 g) of air-dried soil prior to column packing, except for the biochar-based biofertilizer, which was applied as an aqueous suspension (20 g L⁻¹ of tap water) and mixed uniformly with the soil before packing. This approach was adopted to promote homogeneous amendment distribution throughout the soil profile. The soil-amendment mixtures were packed into the columns in small increments with gentle tapping after each addition to achieve consistent packing among experimental units. The packed soil occupied approximately 7–8 cm of the 15-cm column (mean packed height = 7.4 cm), leaving sufficient headspace for application of controlled hydraulic boundary condition during the leaching experiment. Clay powder was incorporated at rates ranging from 1 to 10% (w/w). Compost was applied at three rates: 4% (6 g, C), 8% (12 g, 2C), and 12% (18 g, 3C). Biofertilizer-char was applied at 0.2% (0.3 g, B), either alone or in combination with 4% compost (CB). An unamended soil served as the control. All treatments were replicated three times.

Saturated hydraulic conductivity and bulk density

Saturated hydraulic conductivity (*K_s*) was determined under steady-state conditions using Darcy’s law³⁷ shown in Eq. (1):

$$K = \frac{Q \cdot L}{H \cdot A \cdot T}$$
 (1)

where *Q* is the volume of percolated water (cm³), *L* is the soil column height (cm), *H* is the hydraulic head (cm), *A* is the column cross-section area (cm²), and *T* is the collection time (h).

Prior to measurement, soil columns were saturated overnight. Percolated water was collected at 15-min intervals, and hydraulic conductivity was calculated under saturated conditions.

Bulk density was determined as the ratio between the oven-dried sample mass to its total volume.

Parameter	Unit	Compost	Biofertilizer-char	Parameter	Unit	Compost	Biofertilizer-char
LOI*	%	10.24	–	Mn	ppm	432	155
CaCO ₃ [–]	%	–	–	Zn	ppm	42.6	49.8
pH (1:2.5)		8.23	4.54	Ca ⁺⁺	meq L ⁻¹	15	1.5
E.C	dS m ⁻¹	2.39	0.593	Mg ⁺⁺	meq L ⁻¹	9	1
N ⁺	%	1.34	5.8	Na ⁺	meq L ⁻¹	22.8	41.4
P ⁺	%	0.14	0.174	K ⁺	meq L ⁻¹	0.96	16.2
K ⁺	%	0.74	0.1	HCO ₃ [–]	meq L ⁻¹	2	0.5
Cu ⁻²	ppm	85.4	55	Cl [–]	meq L ⁻¹	27	7.4
Fe ⁺²	ppm	14000	438.8	SO ₄ ⁻²	meq L ⁻¹	18.76	9.4

Table 1. Chemical characteristics for the used natural-based soil amendments (compost and biochar-based microbial fertilizer) in the study. * LOI loss-on-ignition.

Soil water constants and moisture retention characteristics

All soil moisture data presented, analyzed, and discussed in this study are expressed on a gravimetric basis. Saturation percentage (SP) was determined using the saturated paste method and expressed as % (w/w). Soil water content at field capacity (FC) and permanent wilting point (PWP) were determined using the pressure plate method at 0.33 bar and 15 bar respectively. Available water was calculated as the difference between FC and PWP. For the determination of soil moisture retention characteristics, saturated samples were equilibrated on ceramic pressure plates at matric potential of 0.01, 0.10, 0.33, 0.66, 1.00, and 1.50 bar following the procedure of³⁸. Bulk density was determined by an accredited external laboratory for the same samples used in the determination of soil water constants and was calculated as the ratio of oven-dried soil mass to total sample volume. Equilibrium water contents were measured gravimetrically and used to construct the soil moisture retention curves.

Experiment II: effect of soil amendments on water discharge and nitrogen leaching

Experimental design

Experiment II focused on amendments intended to enhance water retention and plant available water while evaluating their implications for nitrogen transport. Therefore, compost, biofertilizer-char, and their combinations were selected for inclusion in the leaching experiment, while clay powder was not considered further in this phase of the study. Factorial soil columns experiment was conducted using acrylic columns prepared as described above. The following soil amendment treatments were evaluated: control soil only, 4% compost (C), 4% compost + biofertilizer-char (CB), 8% compost (2C), and 12% compost (3C).

Each treatment was tested under two urea application levels: 50% and 100% of the recommended nitrogen dose. All treatments were replicated three times. The 50% and 100% urea levels were selected to represent reduced and full recommended nitrogen fertilization scenarios, respectively, consistent with fertilization rates evaluated previously under field conditions³⁰. This design allowed assessment of whether amendment-induced changes in soil physical properties influenced nitrogen leaching across contrasting levels of nitrogen availability.

Column operation and leachate collection

Prior to leaching, soil columns were saturated to establish uniform initial hydraulic conditions among treatments. The subsequent water application was designed as a standardized saturated-flow leaching event to isolate amendment effects on drainage behavior and nitrogen transport under controlled hydraulic conditions rather than to simulate field irrigation practices. During the leaching event, a constant ponded water head was maintained above each column to provide an identical hydraulic driving force so that differences in water flow and nitrogen transport could be attributed primarily to amendment-induced changes in soil physical properties. Columns were placed overnight in beakers containing 50 mL of distilled water, a volume determined in preliminary tests to be sufficient for complete capillary saturation of the packed soil column. Capillary wetting was used to achieve uniform moisture distribution while minimizing soil disturbance and air entrapment compared with direct surface flooding. Leachates were collected one hour after drainage initiation to quantify nitrogen losses associated with the initial drainage event following the standardized saturated-flow leaching event.

Nitrogen analysis and leaching calculations

Total nitrogen in leachate samples was determined using the Kjeldahl method. A 10 mL aliquot of leachate was alkalinized with sodium hydroxide to release ammonia, which was subsequently distilled, trapped in boric acid solution, and quantified by titration with standardized hydrochloric acid using a mixed indicator.

The percentage of leached nitrogen was calculated as shown in Eq. (2):

$$\text{Leached N (\%)} = \frac{\text{Leached N}}{\text{Total N added}} \times 100 \quad (2)$$

Total nitrogen added represented the total nitrogen pool present within each column and included contributions from the soil, amendments, and applied urea. Therefore, the calculated percentage reflects the fraction of the total available nitrogen pool lost through leaching rather than the fraction of fertilizer-N lost. Because compost amendment treatments contain larger nitrogen pools, percentage losses should be interpreted together with absolute nitrogen losses.

Statistical analysis

Descriptive statistics (mean and standard deviation) were calculated for each treatment combination. Data distributions were initially assessed using the Shapiro–Wilk test. Variables that deviated from normality were $\log(x+1)$ -transformed prior to analysis. Two-way analysis of variance (ANOVA) was used to evaluate the main and interactive effects of soil amendment and urea rate. Treatment means were compared using Tukey's HSD test at $p < 0.05$. Following model fitting, ANOVA assumptions were evaluated using Shapiro–Wilk tests of model residuals, Levene's test for homogeneity of variances, and visual inspection of residual-versus-fitted and normal Q-Q plots. Variables requiring transformation were reassessed after transformation to verify that model assumptions were adequately satisfied.

Simple linear regression was used to examine relationships between leached nitrogen and total nitrogen applied, as well as between leached nitrogen and leachate volume. A multiple regression model was fitted to evaluate the combined effects of total nitrogen and leachate volume on nitrogen leaching, with multicollinearity assessed using variance inflation factors (VIF).

All statistical analyses were performed using R software (version 4.5.1; R Core Team, Vienna, Austria). Descriptive statistics were calculated using the *dplyr* package, two-way ANOVA was conducted using the base

stats package, multiple comparisons were performed using Tukey's HSD test implemented in the *agricolae* package, and variance inflation factors (VIF) were calculated using the *car* package. Figures were generated using the *ggplot2* package. All data and analysis scripts are publicly available in a Mendeley Data repository (<https://doi.org/10.17632/6tmfwzjb2v.1>).

Results

Effect of soil amendments on physical properties of sandy soil

The tested soil particle size distribution analysis confirmed a sandy texture composed of 99.34% sand, 0.65% clay, and 0.01% silt. Clay powder analysis indicated kaolinite as the dominant mineral of the powder (61.5%), with quartz (10.5%), microcline (26.1%), and trace accessory minerals (Figure S3). The organic amendments differed markedly in nitrogen concentration, with biofertilizer-char containing substantially greater nitrogen than compost (Table 1). Compost, biofertilizer-char, and their combinations significantly altered key soil physical parameters ($p < 0.05$), including saturation percentage (SP), soil water content at field capacity (FC) and at permanent wilting point (PWP), available water and bulk density, whereas saturated hydraulic conductivity (K_s) was not significantly affected (Table 2).

Saturation percentage ranged from 18.6% in the low-compost treatment (4% compost) to 20.8% in the high compost treatment (12% compost). Field capacity was significantly greater in the biofertilizer-char alone and with the compost + biofertilizer-char combination (13.8–14.1%) than in the compost only treatments. Increasing compost application rates beyond 4% reduced field capacity to 11.2–11.4%. Available water showed a similar response, with the greatest values observed in the 4% compost treatment, biofertilizer-char, and the compost + biofertilizer-char treatments (10.3–10.4%), while the higher compost rate (8% and 12%) resulted in significantly lower available water contents (7.6–8.3%).

Permanent wilting point increased under biofertilizer-char treatment (3.8%) and compost + biofertilizer-char treatment (3.5%), indicating greater water retention under dry soil conditions. In contrast, compost-only treatments, particularly higher rates, showed lower PWP (2.73–3.2%). Bulk density decreased significantly in compost + biofertilizer-char amended soils (1.62 g cm⁻³) relative to 4% compost (1.64 g cm⁻³), reflecting differences in structural packing and organic matter incorporation.

Despite numerical variation (14.2–21.4 cm h⁻¹), K_s did not differ significantly among organic treatments, indicating that changes in water retention occurred largely independently of saturated flow capacity.

In contrast, clay powder application had no significant effect on SP, FC, PWP, and available water (Table S3), but significantly reduced K_s with increasing application rate (Fig. S4). This indicates selective restriction of macropore flow without improvement in plant-available water.

Soil moisture retention curves (Fig. S5) confirmed these trends. biofertilizer-char and compost + biofertilizer-char treatments maintained higher water contents across the full matric potential range, whereas higher compost rates (8% and 12%) exhibited consistently lower moisture retention, consistent with reduced available water.

Amendment-driven control of water discharge and nitrogen leaching

Although clay powder significantly reduced saturated hydraulic conductivity, it produced only limited changes in plant available water. Therefore, subsequent analysis of nitrogen leaching focused on compost, biofertilizer-char, and their combinations, which showed greater potential to modify water retention characteristics.

Two-way ANOVA showed that soil amendment treatment significantly affected nitrogen leaching ($p < 0.001$), whereas urea application rate exerted a smaller but still significant effect ($p = 0.025$). A significant treatment $\times$ urea interaction was also detected ($p = 0.016$), indicating that the response to urea application varied among amendment treatments (Table 3).

Leached nitrogen ranged from 0.09 to 0.63 mg column⁻¹ across treatments. The greatest losses occurred in the 12% compost treatment (0.61–0.63 mg column⁻¹), followed by 8% compost treatment (0.26–0.38 mg column⁻¹). Intermediate losses were observed in biofertilizer-char and compost + biofertilizer-char treatments (0.17–0.20 mg column⁻¹), while the lowest values were recorded in 4% compost and control treatments (0.09–0.13 mg column⁻¹).

A similar pattern was observed for urea leaching. The 12% compost treatment produced the highest urea losses (22.74–28.79%), while lower values were recorded in the biofertilizer-char, compost + biofertilizer-char, 4% compost, and control treatments (2.83–10.41%). Although increasing urea application from 50 to 100%

Treatment	SP % wt	FC % wt	PWP % wt	Available water % wt	Bulk density g cm ⁻³	K_s cm h ⁻¹
4% compost	18.6 ^d	13.6 ^b	3.20 ^{bc}	10.4 ^a	1.64 ^a	20.6 ^a
Biofertilizer-char	19.4 ^{bc}	14.1 ^a	3.80 ^a	10.3 ^a	1.63 ^{ab}	19.3 ^a
Compost + Biofertilizer-char	19.0 ^{cd}	13.8 ^{ab}	3.50 ^{ab}	10.3 ^a	1.62 ^b	21.4 ^a
8% compost	20.1 ^{ab}	11.4 ^c	3.10 ^c	8.3 ^b	1.63 ^{ab}	21.0 ^a
12% compost	20.8 ^a	11.2 ^c	2.73 ^d	7.6 ^b	1.62 ^{ab}	14.4 ^a

Table 2. Effect of organic amendments on physical properties of sandy soil. Different lowercase letters within each column indicate significant differences among treatments ($p < 0.05$). * SP is saturation percent measured by the saturation paste method, FC soil water content at field capacity, PWP soil water content at permanent wilting point, K_s : saturated hydraulic conductivity.

Soil amendment	Applied urea	Leachate volume (mL)	Leached nitrogen (mg column ⁻¹)	Leached nitrogen (mg kg ⁻¹)*	Leached urea (%)*
8% compost	50	333 ± 29.7 ^{bc}	0.26 ± 0.03 ^b	1.73 ± 0.20 ^b	14.52 ± 0.17 ^a
	100	322 ± 41.6 ^{bc}	0.38 ± 0.01 ^b	2.53 ± 0.07 ^b	22.31 ± 3.98 ^a
12% compost	50	249 ± 42.6 ^c	0.63 ± 0.40 ^a	4.20 ± 2.67 ^a	28.79 ± 7.41 ^a
	100	213 ± 8.7 ^c	0.61 ± 0.20 ^a	4.07 ± 1.33 ^a	22.74 ± 0.33 ^a
Biofertilizer-char	50	486 ± 80.9 ^{ab}	0.17 ± 0.04 ^c	1.13 ± 0.27 ^c	5.75 ± 1.79 ^b
	100	437 ± 4.6 ^{ab}	0.19 ± 0.03 ^c	1.27 ± 0.20 ^c	6.10 ± 3.24 ^b
4% compost	50	378 ± 76.2 ^b	0.09 ± 0.01 ^d	0.60 ± 0.07 ^d	5.55 ± 0.76 ^b
	100	433 ± 106.2 ^b	0.09 ± 0.01 ^d	0.60 ± 0.07 ^d	6.54 ± 1.29 ^b
Compost + Biofertilizer-char	50	418 ± 52.5 ^{ab}	0.19 ± 0.03 ^c	1.27 ± 0.20 ^c	2.83 ± 1.72 ^b
	100	439 ± 141.9 ^{ab}	0.20 ± 0.04 ^c	1.33 ± 0.27 ^c	4.72 ± 3.51 ^b
Control	50	471 ± 175.8 ^a	0.11 ± 0.01 ^d	0.73 ± 0.07 ^d	5.88 ± 3.54 ^b
	100	564 ± 143.1 ^a	0.13 ± 0.01 ^d	0.87 ± 0.07 ^d	10.41 ± 5.49 ^b

Table 3. Effect of soil amendments and urea application rate on nitrogen and urea leaching in soil columns. Values are mean ± SD. * Different lowercase letters within each column indicate significant differences among soil amendment treatments according to Tukey's HSD test ($p < 0.05$). Variables were log-transformed prior to ANOVA. Two-way ANOVA detected significant effects of soil amendment, urea application rate, and their interaction on leached nitrogen. Leached nitrogen expressed as mg column⁻¹ and mg kg⁻¹ represents the same response converted using the fixed soil mass of 0.150 kg per column.

generally increased nitrogen losses, differences among amendment treatments were considerably larger than differences attributable to fertilizer rate.

Leachate volume also varied significantly among treatments. The control treatment generated the greatest discharge, whereas compost-amended soils, particularly at higher application rates, produced lower outflow volumes. However, reduced water discharge did not necessarily correspond to lower nitrogen losses, indicating that nitrogen availability and amendment induced changes in retention processes also contributed to leaching behavior.

Coupled controls of nitrogen leaching by nitrogen availability and water flux

Regression analysis revealed a significant positive linear relationship between leached nitrogen and the initial total nitrogen content of the soil column (Fig. 1). Leached nitrogen increased by 4.72×10^{-4} mg column⁻¹ for each additional mg column⁻¹ of total soil N ($\beta = 4.72 \times 10^{-4} \pm 8.93 \times 10^{-5}$ SE; $p < 0.001$). Initial total nitrogen content explained approximately 38% of the variation in nitrogen leaching ($R^2 = 0.378$), indicating that nitrogen availability was an important but not exclusive determinant of nitrogen losses.

Conversely, leached nitrogen was significantly and negatively related to leachate volume (Fig. 2). For every 1 mL increase in leachate volume, leached nitrogen decreased by approximately 5.25×10^{-4} mg ($\beta = -5.25 \times 10^{-4} \pm 1.33 \times 10^{-4}$ SE; $p < 0.001$). Leachate volume explained 25% of the variability in nitrogen leaching ($R^2 = 0.253$). Although this relationship appears counterintuitive, it reflects the contrasting effects of soil amendments on nitrogen availability and water discharge. Treatments with greater compost inputs generally contained larger nitrogen pools while simultaneously producing lower leachate volumes, resulting in higher nitrogen losses despite reduced drainage.

When both predictors were included in a multiple regression model, total soil nitrogen remained positively associated with leached N ($\beta = 4.15 \times 10^{-4}$, $p < 0.001$), while leachate volume maintained a significant negative effect ($\beta = -4.20 \times 10^{-4}$, $p < 0.001$). The combined model was highly significant ($F_{2,45} = 25.81$, $p < 0.001$) and explained 53% of the total variation in nitrogen leaching ($R^2 = 0.534$), indicating joint control by nitrogen availability and hydrological transport processes. Nevertheless, approximately 47% of the variability remained unexplained, suggesting that additional factors such as pore-scale heterogeneity, water-solute contact time, or nitrogen retention processes may also contribute to nitrogen transport dynamics.

These results demonstrate that nitrogen leaching is not governed by a single factor but emerges from the interaction between nitrogen availability and hydrological transport, both of which are strongly modified by soil amendments.

Discussion

The present study was designed as a controlled mechanistic evaluation of how mineral and organic amendments modify sandy-soil physical structure and how these modifications regulate coupled water and nitrogen loss. The use of repacked soil columns enabled isolation of amendment effects on hydraulic behavior and nitrogen leaching under standardized conditions that would be difficult to achieve under field conditions. Although repacking removes natural soil structural heterogeneity and preferential flow pathways, it provides a controlled framework for identifying mechanistic responses to amendment addition. Although plants were not included in the present experiment, the same amendment treatments were previously evaluated in a companion pot study using the same sandy soil³⁰, where improvements in plant growth, nutrient uptake, water-use efficiency, and crop productivity were observed. However, that study did not include direct measurements of soil water

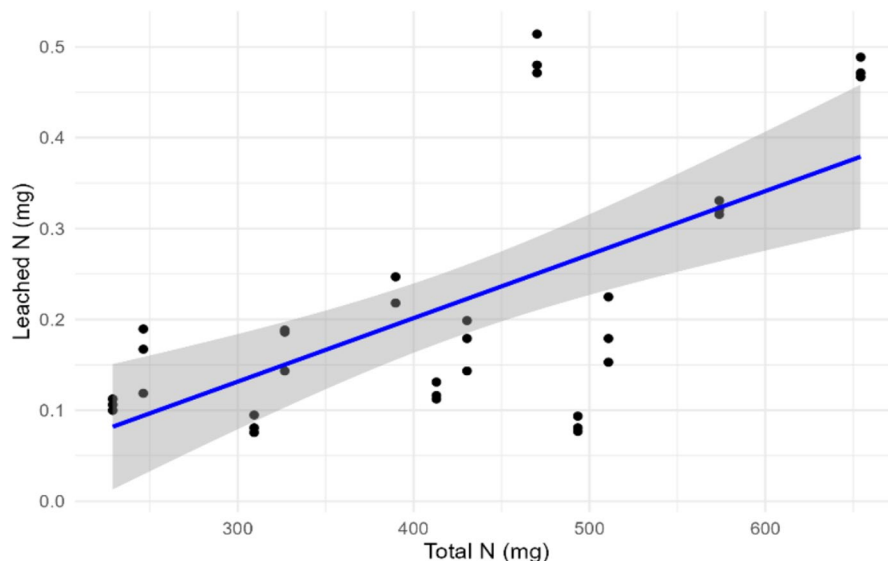


Fig. 1. Relationship between leached nitrogen (mg column^{-1}) and total nitrogen content in the soil column. Solid line represents linear regression; shaded area indicates standard error.

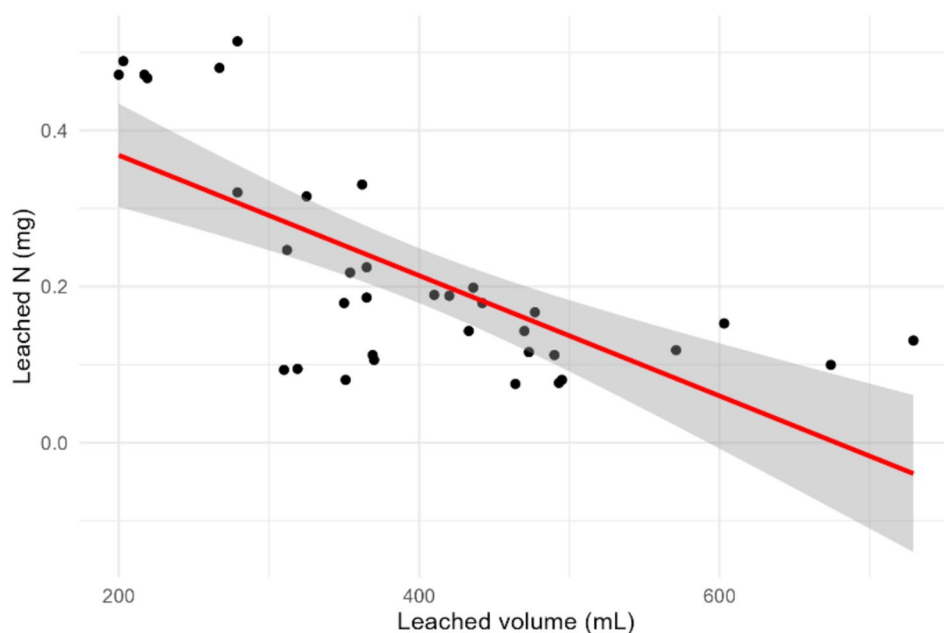


Fig. 2. Relationship between leached nitrogen (mg column^{-1}) and leachate volume (mL). Solid line represents linear regression; shaded area indicates standard error.

content or plant available water and therefore cannot provide direct validation of the hydraulic mechanisms identified here. Rather, the present laboratory investigation provides a mechanistic soil physical and hydrological framework that may explain the improved agronomic responses observed in the companion study. The constant-head leaching test was intentionally adopted to isolate amendment-induced changes in hydraulic behavior under identical boundary conditions. Consequently, the results should be interpreted as mechanistic insights explaining amendment effects on water flow and nitrogen leaching rather than as a direct simulation of irrigation management under field conditions. Nevertheless, caution is required when extrapolating these findings directly to field conditions, where repeated irrigation cycles, natural wetting–drying processes, root activity, biological turnover, and spatial heterogeneity may alter amendment performance and nitrogen transport dynamics over time.

By integrating soil physical responses (Experiment I) with hydrological and nitrogen transport processes (Experiment II), the results demonstrate that amendment effects cannot be interpreted solely in terms of water

retention or nutrient input but must be understood through their influence on pore architecture and flow pathways.

A central outcome is the clear distinction between total water retention and plant-available water. The results further demonstrate that nitrogen leaching was jointly influenced by the initial nitrogen pool present in the soil column and by amendment-induced changes in water movement through the soil profile.

Effect of clay powder on sandy soil physical properties

Clay powder application primarily altered hydraulic behavior by significantly reducing saturated hydraulic conductivity, with minimal impact on plant-available water. This reduction in K_s is consistent with the role of fine particles in obstructing macropores and disrupting pore continuity in coarse-textured soils^{39,40}.

Although clay addition produced consistent numerical increases in saturation percentage, FC, and PWP (Table S2), these changes were not statistically significant. The observed increase in PWP suggests that, under the conditions of the present experiment, a greater fraction of retained water was held at high matric suction, thereby limiting increase in plant available water. Consequently, increases in total water storage were not accompanied by proportional increases in available water.

This trade-off reflects enhanced adsorption and water binding to clay surface, as reported by previous studies^{20,41}. Thus, while clay amendments effectively restrict rapid drainage, their capacity to improve functional water availability in sandy soils remains limited. These findings reinforce the importance of distinguishing between hydraulic restriction and effective water availability when evaluating inorganic amendments, and distinction that has also been emphasized in previous studies of fine-textured amendments in coarse-textured soils^{42,43}.

Comparative performance of organic and inorganic amendments

Clear contrasts appeared between organic amendments (compost and biofertilizer-char) and the inorganic clay amendment in terms of their effects on soil water dynamics. Organic amendments modified soil physical behavior primarily through improvements in aggregation, pore-size distribution, and organic binding, while clay acted mainly through textural alteration^{18,41}.

Compost and biofertilizer-char increased FC and available water without a proportional rise in PWP, resulting in net improvement in plant-available water. This effect was most evident in the combined compost + biofertilizer-char treatment, which achieved a more favorable balance between water retention and availability.

The observed synergy is consistent with previous findings: compost provides labile organic matter that promotes aggregation. The biochar-based biofertilizer treatment may have contributed to improving soil physical conditions through biochar associated structural effects reported in previous studies^{17,44,45}. Biofertilizer-char increased soil water retention across matric potentials, increasing both FC and PWP. Although this did not significantly increase available water compared with compost alone, it contributed to a more stable moisture regime, particularly when combined with compost^{46–48}.

Compost alone improved water retention but showed weaker gains in available water at higher application rates, likely due to rapid decomposition and limited persistence of structural effects. In contrast, biofertilizer-char enhanced structural stability and pore continuity, amplifying the effectiveness of compost when applied together^{49,50}.

Overall, these results demonstrate that organic amendments outperform inorganic clay powder in improving functional water availability, not by maximizing water retention, but by optimizing pore-size distribution.

Effect of amendment rate and excessive compost application

Amendment rate exerted a strong control on soil physical responses and hydrological behavior. Increasing compost applications reduced water discharge and increased total water retention; however, these changes did not consistently improve plant-available water.

At high compost rates (8% and 12%), reduced discharge indicates substantial modification of pore structure, potentially through decreased macropore continuity, increased microporosity, or altered wettability. These changes can promote water retention in pores that are too small to supply plants effectively, thereby reducing functional water availability⁵⁰.

This mechanism explains why combined compost and biofertilizer-char treatments outperformed high compost rates. Biofertilizer-char provided structural stability and persistent pore space, while compost supplied organic binding agents and nutrients, resulting in a more balanced pore system.

This finding highlights that amendment rate is as critical as amendment type, and that excessive organic inputs can disrupt hydraulic balance and reduce functional efficiency.

Effect of soil amendments on water discharge

Water discharge patterns closely reflected changes in soil physical properties. The control treatment exhibited the highest discharge, reflecting rapid gravitational flow typical of sandy soils. Organic amendments, particularly compost and compost + biofertilizer-char treatments, significantly reduced discharge, indicating enhanced water retention within the soil matrix^{51,52}.

This reduction is attributed to increased organic matter content, improved aggregation, and a shift toward finer pore fractions. These structural changes reduce effective hydraulic conductivity and slow water movement, confirming that soil physical modification is a primary control on water flux in sandy systems.

Effect of soil amendments on nitrogen leaching

Nitrogen leaching patterns were strongly coupled with hydrological behavior, confirming that nutrient transport in sandy soil is primarily flow driven.

The control treatment exhibited high nitrogen losses due to limited retention capacity. Compost applications reduced nitrogen leaching at moderate rates, whereas excessive compost application rates (8% and 12%) resulted in the greatest nitrogen losses. Two mechanisms likely contributed to this response. First, higher compost additions increased the total nitrogen pool present within the soil column, thereby increasing the quantity of nitrogen available for potential loss (sources effect)^{53,54}. Second amendment-induced modifications of soil pore structure altered water movement through the soil profile, influencing nitrogen transport and leaching dynamics (transport effect)^{55,56}. These findings suggest that nitrogen losses depend not only on amendment-derived nitrogen inputs but also on how amendments modify hydrological transport processes.

At moderate application rates, reduced nitrogen loss can be attributed to combined mechanisms, including physical retention, microbial immobilization, adsorption onto biochar surfaces, and increased residence time of soil water^{17,57}. Similar reduction in nitrogen loss has been reported for biochar-based microbial fertilizer systems, although the relative contribution of biochar and microbial processes could not be distinguished in the present study⁵⁸.

Increasing urea application rate led to higher absolute nitrogen losses, but the effect was limited and not proportional. This reflects the inherently low nitrogen retention capacity of sandy soils, where excess nitrogen rapidly exceeds retention thresholds and is lost via leaching^{59–61}.

These results demonstrate that nitrogen leaching is not governed solely by fertilizer input, but by the interaction between nitrogen availability and amendment-induced changes in water flow.

Implications for sandy soil management

The findings demonstrate that improving soil physical structure is central to enhancing water and nutrient retention in sandy soils. Organic amendments, particularly the combination of compost and biochar-based microbial fertilizer, provided the most effective balance between water retention, plant-available water, and nitrogen retention.

Clay powder effectively reduced hydraulic conductivity but did not improve functional water availability, limiting its agronomic value. In contrast, organic amendments improved soil function through structural modification, though their effectiveness depended strongly on application rate.

Overall, the results indicate that amendment selection and application rates are equally important for improving sandy soil function. Among the tested treatments, the combination of compost and biofertilizer-char provided the most favorable balance between water retention, plant available water, reduced water discharge, and nitrogen conservation. In contrast, increasing compost application from 4% to 8–12% did not provide proportional benefits and was associated with greater nitrogen losses. These findings agree with previous studies, which suggest that moderate organic amendment rates, particularly when combined with biofertilizer-char, may represent a more effective management strategy than high compost inputs alone for improving irrigation efficiency and reducing nutrient losses especially in arid and semi-arid regions where agricultural production depends on efficient irrigation management^{62–64}.

Study limitations and future perspectives

The present study was designed as a controlled laboratory investigation to identify the mechanisms through which mineral and organic amendments influence soil hydraulic behavior and nitrogen leaching in sandy soils. While this approach allowed amendment effects to be evaluated under standardized conditions, several limitations should be acknowledged.

First, the biochar-based biofertilizer treatment combined biochar and microbial inoculants within a single amendment. Consequently, the relative contributions of biochar-mediated physical effects and microbial process could not be distinguished from the present dataset. Although the observed responses reflect the integrated performance of the amendment as applied in practice, future studies incorporating separate biochar-only and inoculant-only treatments would provide a clearer understanding of their individual effects on soil hydraulic properties and nitrogen dynamics.

Second, the relatively small column dimensions used in this study may have introduced wall effects that influenced water flow pathways, hydraulic conductivity estimates, and leachate volumes. In addition, the constant-head saturated boundary condition and controlled laboratory environment do not fully represent the transient irrigation regimes, structural heterogeneity, preferential flow pathways, root activity, or repeated wetting drying cycles that occur under field conditions. Consequently, the magnitude of water and nitrogen losses should not be interpreted quantitatively as field values, although the relative differences among treatments provide mechanistic insight into amendment effects on hydraulic behavior and nitrogen leaching.

Third, total nitrogen in leachate samples was determined using the Kjeldahl method. Because conventional Kjeldahl digestion does not fully quantify nitrate-N, nitrogen losses may have been underestimated if substantial nitrification occurred during the experimental period. Nevertheless, the same analytical procedure was applied consistently across all treatments, allowing reliable comparison of relative treatment effects on nitrogen leaching.

Despite these limitations, the study provides a mechanistic framework for understanding how amendment-induced changes in soil physical properties influence coupled water and nitrogen transport processes in sandy soils. Future research should combine controlled laboratory investigations with long-term field experiments to evaluate amendment persistence, irrigation responses, nitrogen transformation pathways, and agronomic performance under practical management conditions. In addition, further studies should integrate direct measurements of soil moisture dynamics and plant-available water with agronomic evaluations to verify whether the mechanisms identified under controlled laboratory conditions also operate under practical production field conditions.

Data availability

The datasets generated and analyzed during the current study are publicly available in the Mendeley Data repository: Elbana, Maha (2026), "Data for: Soil physical modification controls water flow and nitrogen retention in amended sandy soils", Mendeley Data, V1, doi:10.17632/6tmfwzjb2v.1

Received: 1 April 2026; Accepted: 4 August 2026

Published online: 17 September 2026

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Funding

This research was supported by the Postgraduate Water Resources Management Scholarship Program, under the Center of Excellence for Water project, in collaboration with USAID and The American University in Cairo.

Declarations

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be considered as a potential conflict of interest.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-65854-1>.

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